Unifying the Gutenberg–Richter Law with Probabilistic Catalog Completeness

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Abstract

Replacing the traditional sharp completeness threshold with a smooth probabilistic transition allows for a more realistic representation of earthquake detection probabilities, acknowledging both missed small events below the completeness magnitude and occasional omissions above it. We propose a probabilistic approach to modeling catalog incompleteness through four candidate augmented Gutenberg–Richter (GR) laws, which incorporate incompleteness into the complementary cumulative frequency–magnitude distribution (CCFMD) using two parameters, m_c the transition magnitude and σ_c, which defines the transition range from incompleteness to completeness. The four augmented GR models are tested within a dedicated validation framework using both synthetic and empirical catalogs. The synthetic catalogs are generated based on the concept of a smooth probabilistic m_c, and the models are rigorously evaluated within a novel framework of correct specification (where each model is fitted to data generated by itself) and misspecification (where a model is fitted to data generated by a different model), using multiple performance evaluation metrics. The GR_{AE} model, which allows for an asymmetry in the convergence to the pure linear GR law for $m > m_c$ relative to the censoring of earthquakes of sizes smaller than m_c, is found to consistently outperform the others in this framework of synthetic and observational tests, providing more robust estimates of seismological parameters (e.g., b value) than existing parameter estimation algorithms, thereby better reflecting realistic physical conditions and observational characteristics. This augmented framework offers three main advantages: (1) unified modeling of incompleteness into the CCFMD, (2) parameters with clear physical and statistical meaning, and (3) the ability to capture nonlinear and asymmetric detection behaviors. It can enhance seismicity models by incorporating small, previously discarded events with appropriate probabilistic weighting, leading to improved seismic hazard assessments and a deeper understanding of earthquake processes.

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Supplemental Material

Introduction

The Gutenberg–Richter (GR) law, first proposed by Gutenberg and Richter (1944, 1954), is a fundamental empirical relationship in seismology that quantitatively characterizes the complementary cumulative frequency–magnitude distribution (CCFMD) of earthquakes in a given region and within a given time window. Mathematically, it is expressed as $\log_{10} N(m) = a - bm$, in which $N(m)$ represents the total number of earthquakes for a selected geographic area and during a specified period with magnitudes $\geq m$, and a and b are empirically derived parameters. Parameter a is an extrapolated estimate of the total number of earthquakes with $m \geq 0$, derived from the linear fit to the observed CCFMD at higher magnitudes. Parameter b quantifies the relative proportion of large to small earthquakes. When normalized by the total number N of earthquakes, it is nothing but an empirical sampling of the underlying complementary cumulative distribution function CCDF(m), describing the probability to observe an earthquake with magnitudes $\geq m$. In other words, $N(m)/N$ converges to CCDF(m) in the limit $N \to \infty$ according to the law of large numbers. The derivative of CCDF(m) with respect to m defines the probability density function PDF(m), which is the likelihood that an earthquake has a specific magnitude m. Although the GR law describes earthquake magnitudes as following an exponential distribution, it translates into a power law

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distribution when expressed in terms of seismic moment or energy, reflecting the scale-invariant characteristics of seismic processes (Bak and Tang, 1989; Sornette and Sornette, 1989; Sornette *et al.*, 1996). Because of its simplicity, robustness, and broad applicability, the GR law constitutes a fundamental cornerstone in seismic hazard assessment, earthquake forecasting, and the study of earthquake physics (Geller *et al.*, 1997; Kagan, 1997; Klügel *et al.*, 2006; Nandan *et al.*, 2022; Mizrahi *et al.*, 2024; Li, Zhang, and Sornette, 2025). Accurate characterization of the CCFMD, or the frequency–magnitude distribution (FMD), is therefore critical not only for probabilistic seismic hazard analyses but also for gaining insights into the physical processes underlying earthquake occurrence and magnitude distribution.

In empirical earthquake catalogs, deviations from the expected linear shape of the GR law plotted on a logarithmic scale are commonly observed at lower magnitudes. Numerous studies have proposed adjusted GR models to characterize these deviations and have explored their physical underpinnings through the lenses of source physics and earthquake fractal characteristics. For instance, Aki (1987) observed that the departure of the empirical CCFMD from the GR law for earthquakes below magnitude 3 coincides with the widely observed deviation occurring at the moment-corner frequency from the $1/\omega^2$ spectrum of seismic waves expressing self-similarity. The corner frequency, characterizing the transition in the seismic-wave spectrum, corresponds directly to a physical scale reflecting earthquake source properties. Several studies have introduced modified GR models to account for observed deviations in earthquake magnitude distributions, employing nonextensive statistical mechanics, a generalization of classical statistical mechanics suitable for complex systems with long-range interactions and fractal structures. For instance, Sotolongo-Costa and Posadas (2004) proposed the fragment-asperity interaction model, which incorporates the irregular geometry of fault planes and the fragments filling the gaps between them, suggesting that these fragments play a significant role in earthquake dynamics. An energy distribution function, which gives the GR law as a particular case, is analytically deduced by them based on the fragment size distribution function derived from a nonextensive formulation. Silva *et al.* (2006) revisited this model within the framework of Tsallis nonextensive statistics, proposing a new GR distribution function that differs from the original. Similarly, Darooneh and Mehri (2010) employed the q-exponential function, motivated by nonextensive statistical mechanics, to propose a modified GR relationship accommodating such deviations. More recently, da Silva (2021) introduced a κ-generalized GR law based on the κ-probability distribution, derived from Kaniadakis statistics.

However, interpretations linking such deviations directly to intrinsic earthquake dynamics may be premature, as these analyses often overlook the critical issue that departures from the GR law are more likely attributable to observational biases and incomplete earthquake catalogs rather than fundamental physical processes. The detectability of earthquakes is influenced by multiple factors, including insufficient station coverage (e.g., Li *et al.*, 2023), high ambient noise levels (e.g., Rydelek and Sacks, 1989), and the intentional configuration of seismic networks, which are typically designed to reliably record earthquakes above a certain magnitude threshold. A temporary increase in the completeness magnitude following large mainshocks is a well-documented phenomenon, often attributable to overwhelmed processing systems (e.g., Zhuang *et al.*, 2017). This demonstrates the limits of routine detection. Moreover, operational and procedural constraints, such as the selection criteria used by agencies like the U.S. Geological Survey (USGS) for publicly releasing events, mean that many smaller earthquakes are intentionally excluded (Andrew Michael, written comm., 2025), further distorting the apparent earthquake size distribution. With the advent of artificial intelligence techniques in seismic event identification and detection, the quality of earthquake catalogs has markedly improved (Kong *et al.*, 2019; Mousavi and Beroza, 2022, 2023; Jia and Zhou, 2024; Kubo *et al.*, 2024). Recent studies show that, in high-resolution catalogs, the magnitude at which deviations from the GR relationship appear has shifted to significantly lower values (e.g., Zhou *et al.*, 2021; Ding *et al.*, 2023; Li *et al.*, 2024; Si *et al.*, 2026). This suggests that the observed deviations in the CCFMD or FMD are still primarily associated with missing earthquake records, underscoring the importance of considering catalog completeness in seismic analyses (Mignan and Woessner, 2012; Li *et al.*, 2023; Wang *et al.*, 2025).

The catalog completeness magnitude, denoted as m_c, is the magnitude above which all earthquakes in a region are assumed to be detected reliably (Rydelek and Sacks, 1989; Mignan and Woessner, 2012). However, this completeness threshold exhibits spatial and temporal variations influenced by factors such as changes in the seismic network, environmental noise, and regional seismicity characteristics (e.g., Mignan *et al.*, 2013; Feng *et al.*, 2022; Li *et al.*, 2023). Most seismological models that utilize the GR law assume a fixed m_c, applying the GR law only above this threshold under the presumption of complete data for $m \geq m_c$. Typically, the catalog's incomplete portion ($m < m_c$) and complete portion ($m \geq m_c$; the GR law) are treated separately (e.g., Woessner and Wiemer, 2005; Mignan, 2012; García-Hernández *et al.*, 2019). More recent innovations have refined this approach: for example, the b-positive estimator introduced by van der Elst (2021) and its improved variant, b-more-positive (Lippiello and Petrillo, 2024), cleverly circumvent explicit m_c selection using only positive magnitude differences between successive earthquakes. These methods are designed to mitigate the influence of time-varying incompleteness, yet they implicitly retain a binary view of detectability. This binary threshold approach oversimplifies the complexities of real-world seismic monitoring, which is more accurately

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characterized by a gradual transition, from undetectability of very small earthquakes, to partial observability of intermediate events, and near-certain detection of larger earthquakes. This binary approach may lead to inaccuracies in seismic hazard assessments and statistical parameter estimations. Therefore, it is crucial to develop models that account for the probabilistic nature of earthquake detection and the dynamic variations in catalog completeness. Such models would provide a more accurate representation of the FMD across the entire spectrum of earthquake magnitudes, leading to improved seismic hazard assessments and a deeper understanding of earthquake processes.

Several probabilistic frameworks for estimating m_c have been developed to overcome the limitations of fixed-threshold approaches, allowing for a more realistic representation of detection capabilities across varying magnitudes:

1. Probabilistic magnitude of completeness (PMC) method: Introduced by Schorlemmer and Woessner (2008), the PMC method estimates earthquake detection probabilities by analyzing empirical data, including phase picks and station metadata. Although this approach offers a graded understanding of detection capabilities, it requires extensive datasets beyond standard earthquake catalogs, such as detailed station information and attenuation relations, making its application complex and data-intensive. In addition, this method is challenging to integrate with other statistical seismological models, particularly the GR model, limiting its use primarily to assessments of catalog data quality and network monitoring capabilities.
2. Ogata and Katsura (1993) model (OK1993): This model employs a cumulative normal distribution function (CDF), as recommended by Ringdal (1975), to represent the monotonically increasing probability of earthquake detection with magnitude and has clear physical interpretations related to instrument sensitivity and background noise. This function is then multiplied by the classical GR PDF to construct a new magnitude distribution model (Ogata and Katsura, 1993, 2006). The model requires estimating parameters μ (the magnitude at which earthquakes are detected with 50% probability) and σ (the size of magnitude changes over which detection transitions from partial to complete), facilitating straightforward data fitting. Ogata and Katsura (1993) also observed that, when earthquakes below a certain threshold magnitude are excluded, a logistic sigmoid function provides a more realistic representation of the detection rate, with the completeness magnitude implicitly defined as $m_c = \mu + n\sigma$, in which n denotes the confidence level (e.g., $n = 0$ for 50% detection, $n = 1, 2, 3$ for approximately 84%, 97.7%, and 99.9% detection, respectively). However, the assumption of a normal distribution imposes mirror symmetry in the detection probability curve, implying that the probability increases toward 1 for magnitudes much greater than μ at the same rate it decreases to 0 for magnitudes much smaller than μ. This symmetry may not align with real-world observations, where detection probability often drops off steeply for smaller magnitudes but increases more gradually toward certainty for larger magnitudes. This imposed mirror symmetry could reduce the model's physical realism, as actual detection probabilities at lower magnitudes may exhibit more complex decay behaviors that a normal CDF cannot accurately capture. Furthermore, the approach of Ogata and Katsura (1993) adopts a multiplicative or separable parameterization, effectively treating the determination of catalog completeness as independent from the statistical modeling of the GR distribution. This separation assumes the absence of coupling between the functions and parameters governing detection and those defining earthquake occurrence rates, preventing a unified treatment of the two constructions that are likely to be coupled. Although the assumed independence provides a natural and tractable starting point, it limits the model's ability to capture potential interdependencies in how catalog imperfections impact the statistical determination of the frequency of earthquakes. Recognizing the nature of the independence assumption in OK1993 makes it well-suited for improvement and generalization.
3. Kijko and Smit (2017) model: This model utilizes a power law function to represent the magnitude detection probability, which is then multiplied by the classical GR distribution to form a new distribution model. This approach accounts for the long tail characteristics of detection probability at lower magnitudes and features a flexible parameter structure, with parameters α and γ capturing diverse detection probability behaviors, theoretically accommodating a broader range of empirical data patterns. However, the parameters α and γ suffer from ambiguous interpretations and lack clear physical meanings, an issue also evident in modified GR relationships proposed by Sotolongo-Costa and Posadas (2004), Silva *et al.* (2006), Darooneh and Mehri (2010), and da Silva (2021). Although power law functions offer significant flexibility, the selection of γ may lack empirical justification. In addition, the strong correlation between α and γ (collinearity) can lead to instability and excessive sensitivity in parameter estimation. Similar to OK1993, this model employs a multiplicative approach that artificially separates completeness from the magnitude distribution, rather than capturing parameter interdependencies within a unified framework. As a result, it lacks a cohesive, concise, and intrinsically integrated structure that reflects the potentially coupled nature of the measure of earthquake frequency and detection processes.

In light of these limitations, we propose a novel enhanced GR model that offers the following advantages: (1) a unified

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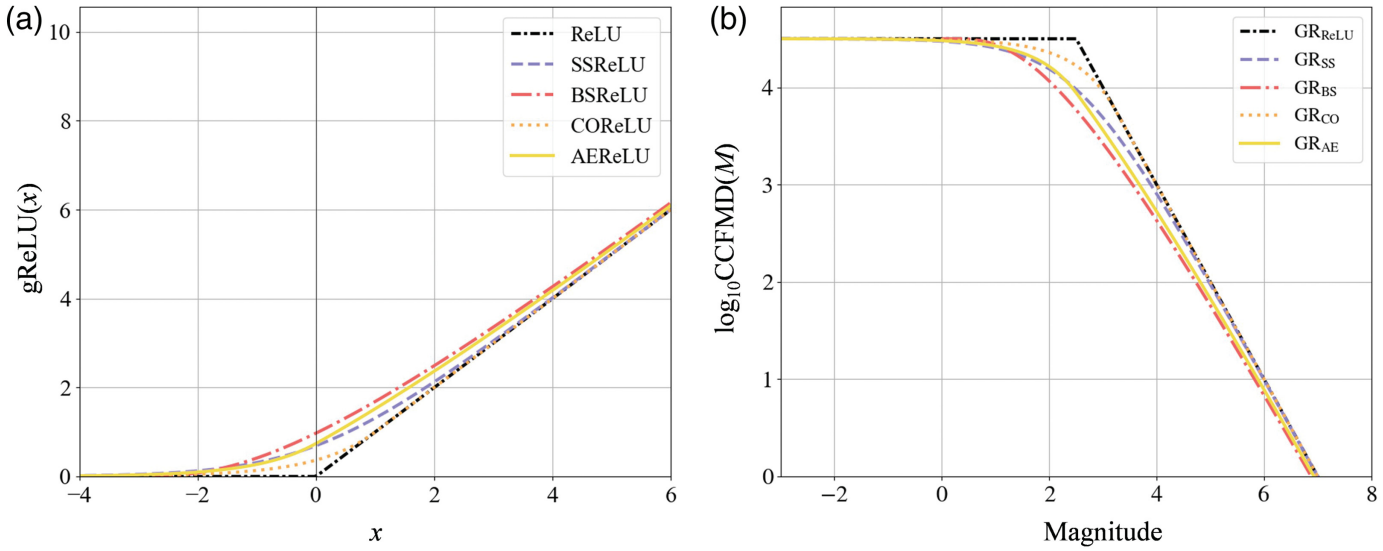


Figure 1. (a) Plots the four candidate gReLU(x) functions. For reference, the classical Rectified Linear Unit, ReLU(x), is shown as a black dashed–dotted line. (b) The corresponding complementary cumulative frequency–magnitude distributions (CCFMDs) generated by each function. The horizontal axis in panel (b) is the earthquake magnitude (m), which relates to the normalized variable (x) in panel (a) via $x = (m - m_c)/\sigma_c$. All plots use the following parameter values, which are informed by typical empirical observations in Table 1: a = 7.0, b = 1.0, m_c = 2.5, σ_c = 0.75, and ζ = 0.35. The curves for the GR_{BS} and GR_{AE} functions are adapted from Wang *et al.* (2025). GR, Gutenberg–Richter.

framework that naturally combines catalog completeness and magnitude distribution within a single expression; (2) parameters with clear physical definitions and intuitive statistical interpretations; and (3) the capability to capture the nonlinear and asymmetric characteristics of magnitude detection probability, accurately reflecting real-world scenarios such as rapid decay in detection probability and asymmetry at lower magnitudes observed in earthquake catalog data.

Four Augmented Forms of the GR Law Using gReLU Functions

Earthquake catalogs often include incomplete portions, which complicates and may bias their analysis. One key challenge is accurately representing the CCFMD of earthquakes recorded in these catalogs. To address this, we propose an augmented GR law tailored to handle incompleteness, expressed as

$$\mathrm{CCFMD}(m) = 10^{a-b\times \mathrm{G(m)}}, \quad (1)$$

in which

$$G(m) = m_c + \sigma_c \times \mathrm{gReLU}(x), \quad (2)$$

in which $x = (m - m_c)/\sigma_c$. This enhanced model augments the traditional GR law by integrating the determination of the CCFMD with a detailed description of catalog completeness. This foundational framework has been introduced and validated in our related work (Wang *et al.*, 2025), where two of its candidate variants (GR_{BS} and GR_{AE}) were compared against traditional estimators based on the truncated m_c assumption. Unlike standard approaches, which treat completeness as a separate issue or an add-on analysis, this formulation seamlessly incorporates completeness into the model itself. Specifically, it extends the classical parameters of the GR law, the a and b values, by introducing two additional parameters: m_c replaces the completeness magnitude as a transition magnitude around which earthquake detection changes rapidly, and σ_c quantifies the magnitude range over which this transition occurs.

The generalized function gReLU(x) is required to approach gReLU(x) = 0 as $x \to -\infty$ and to converge to gReLU(x) = x as $x \to +\infty$ to recover the standard GR distribution for large magnitudes. We do not consider here the issue of tapering or potential outliers in the GR distribution at very large magnitudes, as discussed in Sornette and Sornette (1999), Kagan (2002a,b), Li, Sornette, Wu, and Li (2025), and Li and Sornette (2025). A specific instance of this function is the rectified linear unit, ReLU(x), a widely used activation function in deep learning that introduces nonlinearity into models (Dayan and Abbott, 2001; Nair and Hinton, 2010). The ReLU(x) function is defined as ReLU(x) = 0 for $x < 0$ and ReLU(x) = x for $x \geq 0$ (Fig. 1a). When gReLU(x) is set equal to ReLU(x), the resulting CCFMD(m) becomes 10^{a-bm} for $m \geq m_c$ and the constant 10^{a-bm_c} for $m < m_c$ (Fig. 1b). This corresponds to the exact GR law for $m \geq m_c$, with no earthquakes considered for $m < m_c$, as CCFMD(m) remains constant in this range. Here, we aim to seek functions that provide a smooth transition between the GR law for $m \geq m_c$ and a gradual decrease in recorded earthquake frequencies for $m < m_c$. To satisfy these requirements,

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the smoothing function must not only converge to 0 as $x \to -\infty$ and to x as $x \to +\infty$, but it must also ensure that CCFMD(m) remains monotonic. Consequently, gReLU(x) must be a monotonically increasing function of x. Below are four gReLU(x) functions that meet these criteria (Fig. 1a):

1. SSReLU: A function that satisfies the above requirements is the standard Softplus function, which is commonly used in machine learning and neural networks (Liu and Furber, 2016; Wiemann *et al.*, 2021). In this study, we refer to it as SSReLU, and it is defined by

$$\mathrm{SSReLU}(x) = \ln(1 + e^{x}). \tag{3}$$

2. BSReLU: Another function that meets all these requirements is inspired by the formula for calculating the price of a call option according to the Black–Scholes model (Black and Scholes, 1973; Bouchaud and Sornette, 1994) in financial economics. In this study, we refer to it as BSReLU, which is defined as follows:

$$\mathrm{BSReLU}(x) = \left(x + \frac{m_c}{\sigma_c}\right)\Phi\left(\frac{\ln[(x\sigma_c + m_c/m_c)]}{\sigma_c} + \frac{\sigma_c}{2}\right) - \frac{m_c}{\sigma_c}\Phi\left(\frac{\ln[(x\sigma_c + m_c/m_c)]}{\sigma_c} + \frac{\sigma_c}{2}\right). \tag{4}$$

   In terms of the variable $x = (m - m_c)/\sigma_c$, this corresponds to

$$\mathrm{BSReLU}\left(\frac{m - m_c}{\sigma_c}\right) = \frac{m}{\sigma_c}\Phi\left(\frac{\ln[(m/m_c)]}{\sigma_c} + \frac{\sigma_c}{2}\right) - \frac{m_c}{\sigma_c}\Phi\left(\frac{\ln(m/m_c)}{\sigma_c} - \frac{\sigma_c}{2}\right).$$

   Here, Φ denotes the standard normal CDF. In the supporting information of our separate study (Wang *et al.*, 2025), we show that two independent stochastic constructions, a lognormal exceedance model and a geometric Brownian motion, lead to the same closed-form expression for the BSReLU function.

3. COReLU: Unlike the previous two continuous functions, we also introduce a piecewise function with a cutoff property, which we refer to as COReLU, defined as follows:

$$\mathrm{COReLU}(x) = \begin{cases} e^{x-1} & x < 1 \\ x & x \geq 1 \end{cases}. \tag{5}$$

   This function is composed of an exponential function and a linear function. Unlike ReLU(x), which consists of two distinct linear segments joined at $x = 0$, COReLU(x) consists of one nonlinear curve and one linear segment joined at $x = 1$. This design ensures a smooth transition of the function while avoiding a step discontinuity at the cutoff point. In other words, COReLU(x) and its first derivative are continuous at $x = 1$.

4. AEReLU: This function allows for an asymmetric exponential convergence to 0 for $x < 0$ and to x for $x > 0$, in which ζ controls the strength of the asymmetry (Wang *et al.*, 2025):

$$\mathrm{AEReLU}(x) = \begin{cases} \frac{e^{x}}{1+\zeta} & x < 0 \\ x + \frac{e^{-\alpha x}}{1+\zeta} & x \geq 0 \end{cases}. \tag{6}$$

   This parameterization again ensures continuity and differentiability at the transition point $x = 0$.

In terms of the variable $x = (m - m_c)/\sigma_c$, in the limit $\sigma_c \to 0^+$, the augmented mapping $G(m) = m_c + \sigma_c \mathrm{gReLU}[(m - m_c)/\sigma_c]$ converges to $m_c + \mathrm{ReLU}(m - m_c) = \max(m, m_c)$, thus recovering the pure GR for $m \geq m_c$ and a constant for $m < m_c$. In summary, the SSReLU function models a symmetric crossover controlled by exponential decay, whereas the BSReLU function employs a symmetric crossover governed by a Gaussian profile. The COReLU function features exponential tapering only on the side of incompleteness. The AEReLU function extends SSReLU by introducing asymmetry through an additional parameter ζ, allowing for an asymmetric crossover characterized by different exponential forms on the two sides of the transition.

By substituting the smoothing function gReLU(x) from equations (3)–(6) into equations (1) and (2), we obtain four distinct augmented GR models. These models are referred to as $\mathrm{GR_{SS}}$, $\mathrm{GR_{BS}}$, $\mathrm{GR_{CO}}$, and $\mathrm{GR_{AE}}$, respectively (Fig. 1b). It can be observed that $\mathrm{GR_{SS}}$, $\mathrm{GR_{BS}}$, and $\mathrm{GR_{CO}}$ share the same degrees of freedom, meaning they are characterized by an identical set of fitting parameters, $\theta = \{a, b, m_c, \sigma_c\}$. $\mathrm{GR_{AE}}$ introduces an additional parameter, expanding its parameter set to $\theta = \{a, b, m_c, \sigma_c, \zeta\}$. It is also worth noting that, by definition, the BSReLU function is only valid when both m and m_c are positive. Therefore, if the catalog to be fitted contains events with magnitudes less than zero, a shift parameter C should be added to both m and m_c in equations (2) and (4). In practice, C is typically chosen as the absolute value of the minimum magnitude in the catalog, so that the shifted catalog has a minimum magnitude of zero. The $\mathrm{GR_{BS}}$ model is then fitted in the shifted magnitude space, and after fitting, both m and m_c are shifted back to recover the parameter estimates in the original magnitude scale. The fitting procedure is detailed in Text S1, available in the supplemental material to this article. Figure 1 illustrates the four gReLU(x) functions and their corresponding CCFMD(m) representations when applied to equations (1) and (2).

Earthquake Catalog Used for Testing

We evaluate the augmented GR model, incorporating the gReLU(x) functions, using both simulated earthquake catalogs and empirical earthquake catalogs. For the simulated earthquake catalogs, events are generated based on PDFs derived from each of the augmented GR models incorporating the four candidate gReLU(x) functions. The PDFs have a minimum

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magnitude of $m_{min} = 0$ and a maximum magnitude of $m_{max} = 8.0$. The upper bound was chosen to cover the full range of magnitudes relevant for regional seismicity studies, whereas the lower bound allows us to explicitly test model behavior in the incomplete part of the catalog. The parameters used are $b^{true} = 1.0$, $m_c^{true} = 2.5$, $\sigma_c^{true} = 0.75$, and $\zeta^{true} = 0.35$, which are representative of typical values observed in real earthquake catalogs (see Table 1), with the total number of earthquakes set to 10^k for $k = 3, 4, 5, 6$, allowing us to systematically assess convergence and uncertainty reduction as a function of sample size. The true a value a^{true} is calculated as $a^{true} = k + 2.5 + 0.75\text{gReLU}(-10/3)$, based on equations (1) and (2). Specifically, $\text{SSReLU}(-10/3) \approx 0.03505$, $\text{BSReLU}(-10/3) \approx 0$, $\text{COReLU}(-10/3) \approx 0.01312$, and $\text{AEReLU}(-10/3) \approx 0.02378$. These setups result in a total of 16 synthetic datasets, as each model generates data for four different values of k. For each combination of model and k, 200 independent datasets are sampled directly from the specified PDF, yielding 200 parallel realizations per group for subsequent performance evaluation and analysis.

The observed earthquake catalogs are selected from four regions in China and two additional regions, California and New Zealand, totaling six study regions (Fig. 2). These empirical catalogs and their regional definitions follow the dataset and preprocessing framework established in Wang *et al.* (2025). In China, the regions include Beijing–Tianjin–Hebei, the southeastern coast, Sichuan-Yunnan, and northern Xinjiang. Their boundaries are defined by the $m_c \approx 1.8$ contour lines from the posterior m_c map generated using the Bayesian magnitude of completeness (BMC) method in Li *et al.* (2023). Data were sourced from the China Earthquake Networks Center (CENC) for the period 1 January 2009 to 18 November 2024. The California catalog, from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat; U.S. Geological Survey, 2017), spans 1 January 2009 to 2 November 2023. Its boundary follows the $m_c \approx 1.8$ contour lines derived from the BMC method in Tormann *et al.* (2014). The New Zealand catalog, from the GeoNet Earthquake Catalog (GNS Science, 2022), covers 1 January 2009 to 12 December 2023. Its boundary is based on Petersen *et al.* (2011), focusing on areas with dense and evenly distributed seismic stations. In this study, earthquakes with focal depths larger than 100 km were excluded from all six study regions. Figure 2 presents the spatial boundaries of the six study regions and the distribution of earthquakes with magnitude ≥ 3 within them, alongside the magnitude–time sequence plots for the respective empirical earthquake catalogs.

Results

Model evaluation results

Tables S1a–S4a show the fitted parameters and performance evaluation metrics of the augmented GR model incorporating the four gReLU(x) functions, applied to simulated catalogs generated by the GR_{SS}, GR_{BS}, GR_{CO}, and GR_{AE} models, respectively, using predefined parameters (see the Earthquake catalog used for testing section for details). Table 1 and Table S5a show the fitted parameters and performance evaluation metrics of the augmented GR models applied to empirical earthquake catalogs from six study regions. To systematically assess model performance, we employ four evaluation metrics, root mean square error (RMSE), coefficient of determination (R-Square), sum of squared errors (SSE), and Akaike information criterion (AIC; Akaike, 2011), to determine which of the four augmented GR models performs best under each scenario. Table S5a also reports the p values for empirical earthquake catalogs. Here, the p value represents the probability that synthetic catalogs generated by the model show a poorer fit than the empirical catalog. Consequently, larger p values indicate that the model provides an adequate fit to the empirical data, whereas smaller p values suggest a poorer fit, potentially warranting model rejection. Tables S1b–S5b indicate the rankings of GR_{SS}, GR_{BS}, GR_{CO}, and GR_{AE} based on these metrics, as presented in Tables S1a–S5a, where a rank of "First" denotes the best-performing model for a given metric.

Uncertainties in different model parameters contribute unequally to the overall model uncertainty. In our framework, m_c and σ_c are generally well-constrained, whereas the asymmetry parameter ζ often exhibits larger uncertainty, in some cases comparable to or exceeding its estimated value (Table 1). This reflects the fact that ζ controls higher-order asymmetry in the completeness transition and is weakly identifiable when the data do not strongly constrain such asymmetry. Importantly, this uncertainty has a limited impact on the estimation of core seismicity parameters, such as the b value, which is primarily constrained by the near-complete, high-magnitude part of the catalog (see e.g., the Robustness and stability of asymptotic b value estimates section).

Figure 3 shows the proportion of times each model ranked as best (first, dark red), good (second, light red), poor (third, light blue), or worst (fourth, dark blue) for simulated and empirical catalogs. In Figure 3, panels (e–i) and (j–o) present alternative views of the same data from Table S5: (Fig. 3e–i) shows, for each metric, how often each model ranks first to fourth across the six study regions, whereas (Fig. 3j–o) shows, for each region, how often each model ranks first to fourth across the five evaluation metrics. Among the four candidate models, GR_{AE} demonstrates the greatest robustness and is therefore recommended as the preferred choice. For detailed analyses and comparisons of all models, we refer to Texts S2–S4. In the cases of correct model specification—cases where each model is fitted to data generated by itself, GR_{AE} exhibits favorable performance (first and second place). In the cases of model mis-specification—cases where a model is fitted to data generated by a different model, GR_{AE} also consistently demonstrates greater robustness, ranking as best or good in at least

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TABLE 1
Fitted Parameters of the Augmented Gutenberg–Richter (GR) Model with Four gReLU(x) Functions Applied to Empirical Earthquake Catalogs from Six Study Regions

		Fitted Parameters				
Region	**Model**	a^{pred}	b^{pred}	m_c^{pred}	σ_c^{pred}	ζ^{pred}
Beijing–Tianjin–Hebei	GR_{SS}	5.48 ± 0.07	0.95 ± 0.03	0.69 ± 0.05	0.32 ± 0.03	–
	GR_{BS}	5.85 ± 0.42	1.03 ± 0.07	0.62 ± 0.43	0.73 ± 0.21	–
	GR_{CO}	5.42 ± 0.04	0.93 ± 0.02	0.64 ± 0.03	0.50 ± 0.03	–
	GR_{AE}	5.56 ± 0.12	0.97 ± 0.04	0.76 ± 0.09	0.38 ± 0.04	0.56 ± 0.11
Southeastern coastal	GR_{SS}	4.97 ± 0.08	0.93 ± 0.04	0.51 ± 0.06	0.28 ± 0.04	–
	GR_{BS}	5.62 ± 0.78	1.05 ± 0.11	0.74 ± 0.62	0.93 ± 0.28	–
	GR_{CO}	4.93 ± 0.06	0.92 ± 0.03	0.46 ± 0.05	0.42 ± 0.05	–
	GR_{AE}	5.06 ± 0.13	0.96 ± 0.05	0.58 ± 0.10	0.32 ± 0.06	0.51 ± 0.10
Sichuan–Yunnan	GR_{SS}	6.48 ± 0.03	0.86 ± 0.01	0.81 ± 0.02	0.21 ± 0.01	–
	GR_{BS}	6.79 ± 0.02	0.92 ± 0.02	1.10 ± 0.01	0.95 ± 0.07	–
	GR_{CO}	6.48 ± 0.02	0.86 ± 0.01	0.81 ± 0.01	0.41 ± 0.01	–
	GR_{AE}	6.72 ± 0.14	0.87 ± 0.01	1.09 ± 0.16	0.39 ± 0.02	0.39 ± 0.72
Northern Xinjiang	GR_{SS}	5.79 ± 0.07	0.89 ± 0.03	1.13 ± 0.05	0.21 ± 0.03	–
	GR_{BS}	6.08 ± 0.09	0.95 ± 0.03	1.36 ± 0.07	0.62 ± 0.09	–
	GR_{CO}	5.78 ± 0.06	0.89 ± 0.02	1.12 ± 0.04	0.41 ± 0.03	–
	GR_{AE}	6.14 ± 0.14	0.92 ± 0.06	1.47 ± 0.08	0.39 ± 0.03	0.11 ± 0.20
California	GR_{SS}	6.39 ± 0.09	1.13 ± 0.03	0.84 ± 0.05	0.35 ± 0.04	–
	GR_{BS}	6.94 ± 0.38	1.23 ± 0.08	1.10 ± 0.31	0.88 ± 0.17	–
	GR_{CO}	6.27 ± 0.05	1.09 ± 0.02	0.76 ± 0.03	0.49 ± 0.04	–
	GR_{AE}	6.55 ± 0.14	1.17 ± 0.04	0.96 ± 0.09	0.41 ± 0.05	0.46 ± 0.07
New Zealand	GR_{SS}	7.16 ± 0.06	1.03 ± 0.02	1.78 ± 0.03	0.24 ± 0.02	–
	GR_{BS}	7.45 ± 0.09	1.09 ± 0.02	1.95 ± 0.05	0.37 ± 0.04	–
	GR_{CO}	7.13 ± 0.04	1.03 ± 0.01	1.76 ± 0.02	0.42 ± 0.01	–
	GR_{AE}	7.63 ± 0.10	1.11 ± 0.02	2.07 ± 0.06	0.39 ± 0.01	0.24 ± 0.16

The parameter estimates for the GR_{BS} and GR_{AE} models are derived from the analyses presented in Wang *et al.* (2025).

70% of cases, with no occurrences of being ranked as worst. These results indicate the superior adaptability of GR_{AE}.

In Figure 3, GR_{AE} stands out by attaining best or good rankings significantly more often than the other three models for the empirical catalogs from the six study regions. The GR_{AE} model demonstrates consistently strong performance across diverse regions (except Beijing–Tianjin–Hebei), affirming the model's flexibility and robustness in capturing realistic earthquake completeness and magnitude distributions. Figures S1–S5 display the PDFs of the estimated values of m_c^{pred}, σ_c^{pred}, a^{pred}, b^{pred}, and ζ^{pred}, respectively, obtained from 200 bootstrap iterations. Figure 4 shows $CCFMD^{obs}(m)$ of the empirical catalogs alongside the fitted $CCFMD^{fit}(m)$ derived using the GR_{AE} model. Parallel results for the other three models are displayed in Figures S6–S8. Figure S9 shows residuals between $CCFMD^{obs}(m)$ and $CCFMD^{fit}(m)$, expressed as $\log_{10} CCFMD^{fit}(m) - \log_{10} CCFMD^{obs}(m)$.

The distinct performance patterns of the candidate models also offer insights into the physical processes governing catalog incompleteness. The similar parameter estimates and uncertainties obtained for GR_{SS} and GR_{CO} (Table 1 and Figs. S1–S4) arise from their shared functional characteristics, namely, an exponential-to-linear transition and an identical high-magnitude linear asymptote. Under our heteroscedastic weighting scheme, the empirical data provide limited leverage to distinguish the subtle differences in their curvature near the

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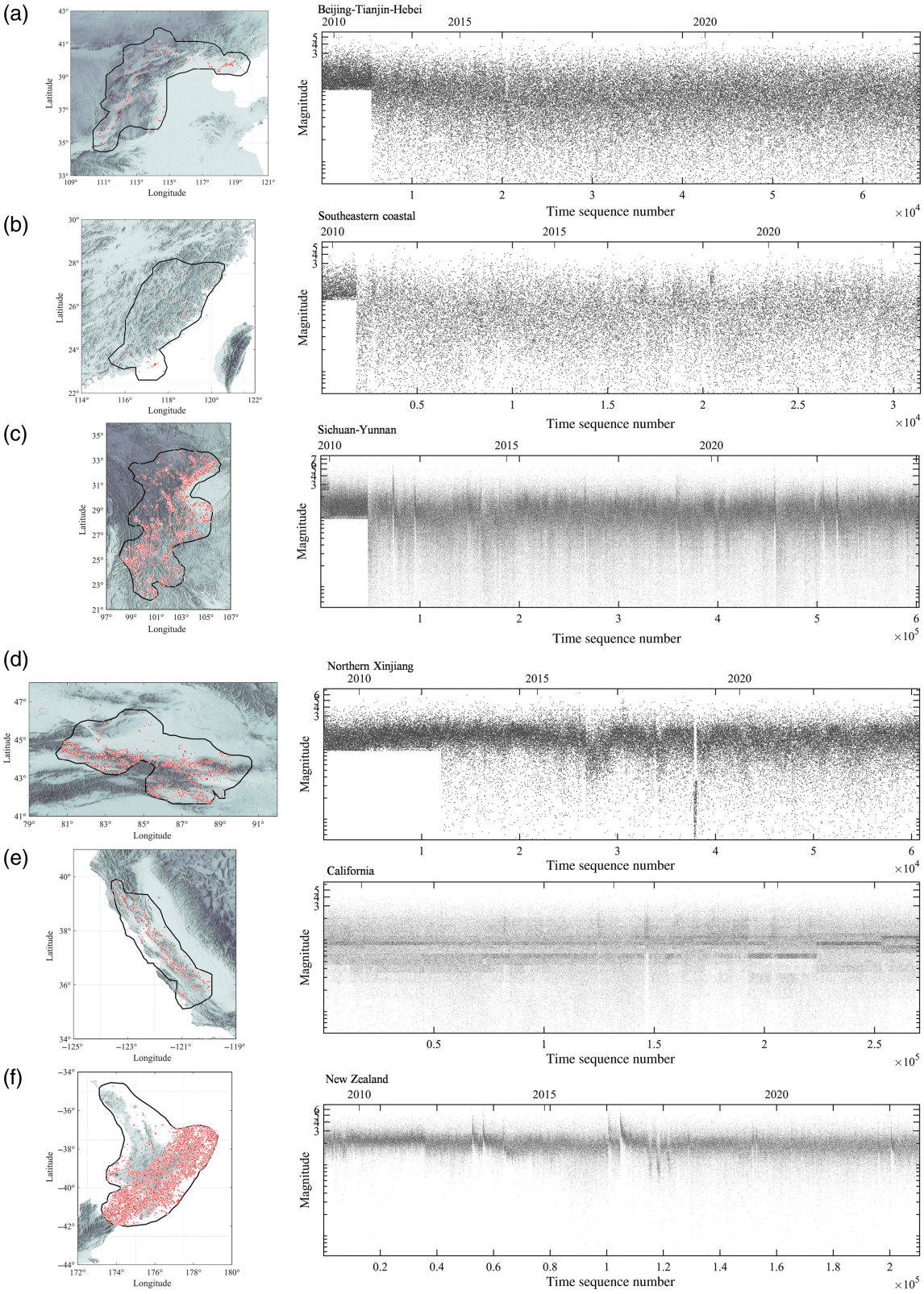


Figure 2. Spatial distribution and magnitude–time sequence of earthquakes in the six study regions: (a) Beijing–Tianjin–Hebei, (b) the southeastern coast, (c) Sichuan–Yunnan, (d) northern Xinjiang, (e) California, and (f) New Zealand. This figure combines maps showing the study area boundaries and the spatial distribution of earthquakes ($m \geq 3.0$) with their corresponding magnitude–time sequence plots. Earthquake magnitudes are perturbed by adding uniformly distributed random errors within the interval [–0.05, 0.05].

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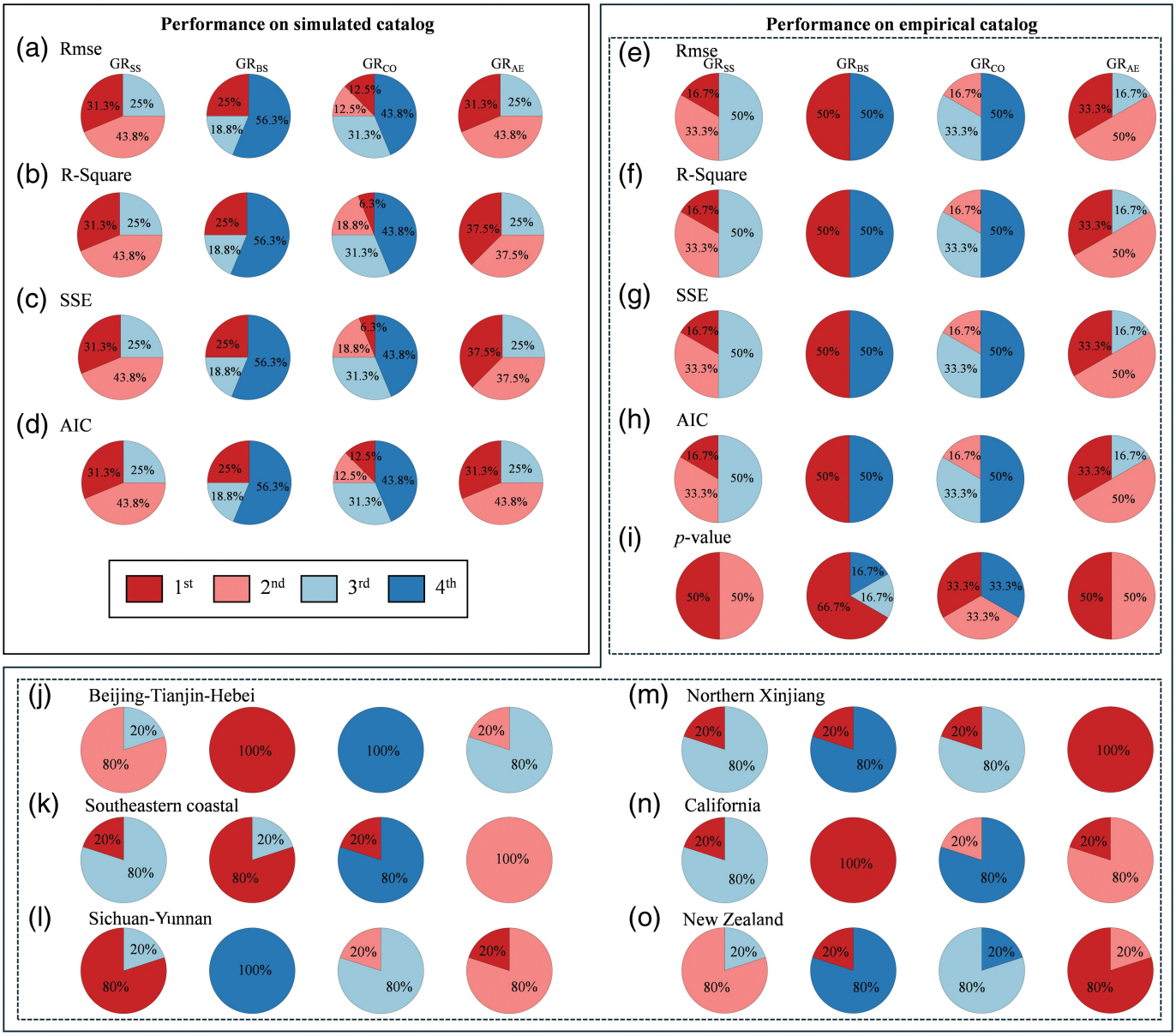
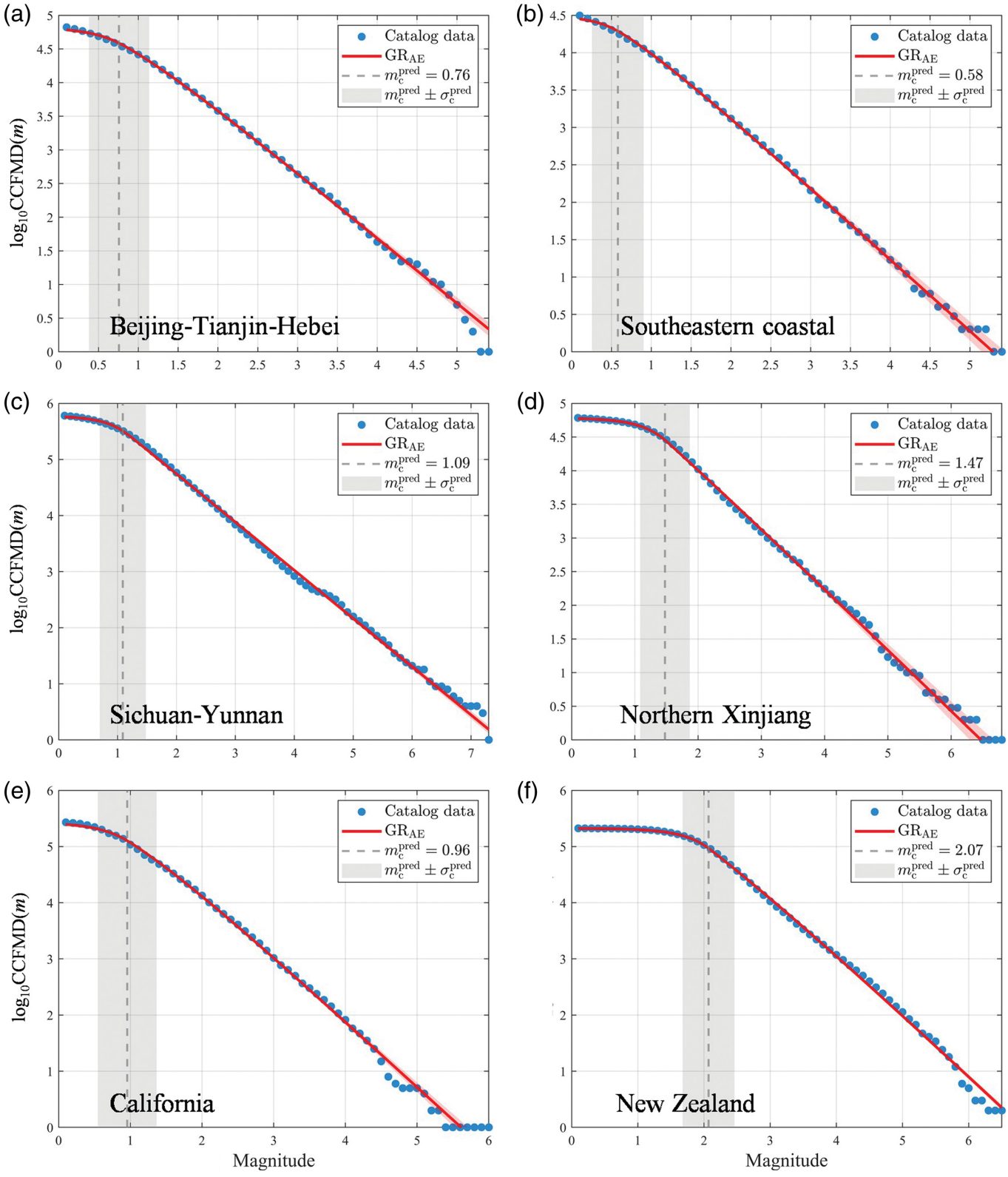


Figure 3. Proportions of times each model (GR_{SS}, GR_{BS}, GR_{CO}, and GR_{AE}) ranked as best (first, dark red), good (second, light red), poor (third, light blue), or worst (fourth, dark blue) across different evaluation settings. Panels (a–d) summarize model rankings based on four metrics (root mean square error [RMSE], R-Square, sum of squared errors [SSE], and Akaike information criterion [AIC]) for simulated catalogs. Panel (e–o) presents rankings based on five metrics (including *p* value) for empirical catalogs, offering two complementary views of the same data from Table S5: panel (e–i) shows, for each metric, how often each model ranks first to fourth across the six study regions, whereas panel (j–o) shows, for each region, how often each model ranks first to fourth across the five evaluation metrics. Detailed rankings and metric values for each model are provided in Tables S1–S5.

transition point. In contrast, the bimodal performance of the GR_{BS} model shown in Figure 3 stems from its strong structural assumptions. It achieves the best fit in regions such as Beijing–Tianjin–Hebei and California, where the empirical completeness transition closely resembles a lognormal exceedance process (Wang *et al.*, 2025). However, this specificity compromises its flexibility, leading to poorer performance in other tectonic settings. This pronounced context-dependence suggests that the GR_{BS} model can serve as a diagnostic tool to identify regions where catalog incompleteness is primarily governed by multiplicative stochastic processes.

In summary, the four augmented GR models exhibit distinct behaviors that reflect different trade-offs between structural assumptions and flexibility. GR_{SS} provides a smooth, general-purpose transition from incompleteness to completeness, resulting in stable and typically intermediate performance across regions. GR_{CO} adopts a piecewise but continuously differentiable form, offering a conservative representation that avoids overfitting while maintaining robustness. In contrast, GR_{BS} is characterized by strong structural assumptions rooted

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in stochastic processes such as lognormal exceedance and geometric Brownian motion; as a result, it tends to display polarized performance, achieving the best fits in regions where these assumptions are met, but performing less well otherwise, thus serving as a diagnostic rather than a universally robust model. GR_{AE} introduces an explicit asymmetry parameter, enabling

Figure 4. CCFMD of the empirical earthquake catalogs from six study regions, along with the fitted CCFMD(m) derived using the augmented GR law with the AEReLU(x) function (GR_{AE}): (a) Beijing–Tianjin–Hebei, (b) the southeastern coast, (c) Sichuan–Yunnan, (d) northern Xinjiang, (e) California, and (f) New Zealand. The red shading represents the estimated uncertainty, obtained from 200 bootstrapping iterations.

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flexible and adaptive modeling of heterogeneous completeness transitions, and consequently demonstrates consistently strong performance across diverse regions. Together, these models span a spectrum from physically interpretable but restrictive formulations to flexible and robust representations, allowing both diagnostic insight and reliable estimation of seismicity parameters.

Seismicity characterization across regions: Insights from a multimodel augmented GR framework

The differences in the b^{pred} values estimated by the four augmented GR models proposed in the present study are often larger than their respective standard deviations (Table 1 and Fig. S2). This indicates that the differences are not negligible and may carry meaningful information. In some cases, such as GR_{SS}, GR_{CO}, and GR_{AE} in the Sichuan-Yunnan region, the estimated b^{pred} values are identical or nearly identical, suggesting that the consistent estimates across models reflect a precise and reliable value of b. To take advantage of this variability while accounting for the differing levels of model reliability (i.e., explanatory power for the observed data), we compute a weighted average of the b^{pred} value PDFs from the four augmented GR models. This yields a unified, weighted estimate of the regional b value, expressed as

$$\ln[\text{PDF}_{\text{weight}}(b^{\text{pred}})] = \sum_{i=1}^{4} \omega_i \ln[\text{PDF}_i(b^{\text{pred}})], \tag{7}$$

in which ω_i is the weight assigned to model i. To calculate these weights, we adopt the AIC in Table S5a, which considers both goodness-of-fit and model complexity by penalizing overparameterization. The AIC-based weight is given by

$$\omega_i = \frac{\exp(-\frac{1}{2}\Delta\text{AIC}_i)}{\sum_{j=1}^{4}\exp(-\frac{1}{2}\Delta\text{AIC}_j)}, \tag{8}$$

with $\Delta\text{AIC}_i = [\text{AIC}_i - \min(\text{AIC})]$. This approach is analogous to Bayesian inference, where a posterior distribution is obtained by combining a prior and a likelihood. Here, the weighted average of multiple PDFs is obtained using a geometric weighting scheme derived from AIC differences. In principle, this procedure can also be applied to other parameter estimates beyond b^{pred} values. The resulting combined distributions represent our most robust and comprehensive parameter estimates.

Figure 5 shows the weighted averages of the b^{pred} value PDFs from the four augmented GR models for the six study regions (see also Fig. S2 and Table 1). Among the six regions, Sichuan-Yunnan and northern Xinjiang yield mean b^{pred} values below 1.0, whereas mean b^{pred} values for the southeastern coast and Beijing–Tianjin–Hebei are closest to 1. In contrast, California and New Zealand exhibit mean b^{pred} values greater than 1. The weighted average PDFs for Sichuan–Yunnan and New Zealand show relatively small standard deviations,

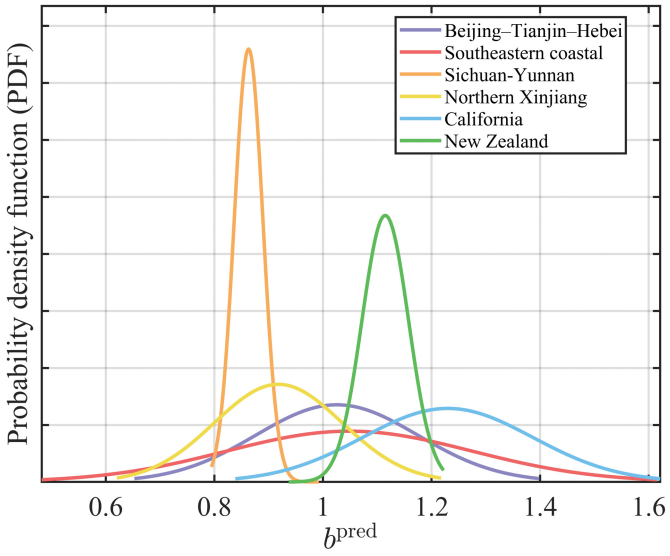


Figure 5. Weighted average of the b^{pred} value probability density functions (PDFs) from the four augmented GR models for the six study regions. Details of the weighting procedure are described in the Seismicity characterization across regions: Insights from a multimodel augmented GR framework section. The individual PDFs of the estimated b^{pred} value are provided in Figure S2 and Table 1.

indicating strong consistency among the estimates from the four models and suggesting that the resulting b^{pred} values in these regions are precise and reliable. In the remaining four regions, the standard deviations are notably larger, reflecting greater variability across models. Our estimated b^{pred} values are generally consistent with previous studies that used traditional approaches such as least-squares fitting, maximum-likelihood estimation (MLE), or data-driven techniques over similar spatial and temporal domains. For example, $b \approx 1$ in the Beijing–Tianjin–Hebei region has been reported by Bi *et al.* (2025), Xiong *et al.* (2019), and Shen *et al.* (2024); similar values for the southeastern coast are noted in Zheng *et al.* (2022). A b value below 1 in the Sichuan-Yunnan region is consistent with findings from Xiong *et al.* (2020), Liu *et al.* (2024), and Zhang and Zhou (2016). In northern Xinjiang, $b < 1$ is supported by Li, Li, *et al.* (2025) and Zhang and Tang (2015), whereas $b > 1$ in California aligns with Kamer and Hiemer (2015) and Scholz (2015). Similarly, elevated b values in New Zealand are consistent with Lu (2016, 2017).

Discussion

The superior performance demonstrated by GR_{AE} model, which incorporates the additional parameter ζ to account for the asymmetric crossover between the pure linear GR law for $m \geq m_c^{\text{pred}}$ and the incomplete part of the catalogs of earthquakes of sizes smaller than m_c^{pred}, underscores the critical importance of explicitly accounting for asymmetry in completeness modeling. Unlike symmetric functions such as GR_{SS}

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and GR_{BS}, or the single-sided exponential tapering exhibited by GR_{CO}, the GR_{AE} model's asymmetry provides enhanced flexibility. This flexibility enables a more accurate and realistic representation of inherently asymmetric detection probabilities across various magnitude ranges, resulting in improved completeness modeling, which should translate into more reliable seismic hazard assessments. Our probabilistic approach offers a significantly more realistic and comprehensive characterization of earthquake catalogs, emphasizing the necessity of integrating region-specific detection attributes into statistical seismic modeling frameworks.

Our framework additionally overcomes the influence of magnitude binning on parameter estimation (e.g., Marzocchi and Sandri, 2003; Marzocchi *et al.*, 2020). Our approach naturally accommodates such discretization, as the augmented GR model is fitted directly to the observed CCFMD rather than relying on differential frequency counts, which are highly sensitive to bin size. Moreover, the probabilistic completeness function effectively smoothes the transition across magnitude bins near m_c, reducing discretization artifacts. Although alternative treatments, such as explicitly modeling rounded magnitudes or adopting continuous-magnitude likelihoods, are possible, we find that the current formulation is robust for catalogs with the standard 0.1 magnitude binning.

In this section, we primarily use the best-performing GR_{AE} model to illustrate the concept of magnitude-dependent catalog completeness probability as defined by our proposed framework, as well as the asymmetric completeness characteristics revealed by the varying ζ^{pred} values across regions. In addition, we compare the b value estimates from the GR_{AE} model, which vary with the catalog cutoff magnitude M_{co}, against those derived from MLE and the b-more-positive estimator. We also discuss the application of this framework to a higher-resolution regional catalog (induced seismicity in Oklahoma, United States). Together, these analyses further substantiate the robustness and reliability of the GR_{AE} model and the broader augmented GR framework.

Magnitude-dependent catalog completeness probability

In practice, the catalog completeness probability $P(m)$, that is, the probability that events of magnitude m or larger are recorded in a given region, can be defined as the ratio between equation (1) and the linear GR law, expressed as

$$P(m) = \frac{10^{a-b\times G(m)}}{10^{a-b\times m}} = 10^{b[m-G(m)]}. \quad (9)$$

Substituting equation (2), along with $x = (m - m_c)/\sigma_c$, into equation (9), we obtain:

$$P(m) = 10^{b\sigma_c[\frac{m-m_c}{\sigma_c} - \mathrm{gReLU}(\frac{m-m_c}{\sigma_c})]}. \quad (10)$$

Thus, by substituting the gReLU(x) expressions from equations (3) to (6) and the corresponding parameter estimates (Table 1) into equation (10), one can compute the completeness probability for any given magnitude m.

Based on the GR_{AE} estimates, we computed completeness probabilities at $m = m_c^{pred} + \sigma_c^{pred}$, $m = m_c^{pred} + 2\sigma_c^{pred}$, and $m = m_c^{pred} + 3\sigma_c^{pred}$ for all six study regions. In Beijing–Tianjin–Hebei, the completeness probabilities at $m = m_c^{pred} + \sigma_c^{pred} = 1.14$, $m = m_c^{pred} + 2\sigma_c^{pred} = 1.52$, and $m = m_c^{pred} + 3\sigma_c^{pred} = 1.90$ are 73.3%, 83.7%, and 90.4%, respectively. The southeastern coastal region performs comparably, with probabilities ranging from 75.5% at $m = m_c^{pred} + \sigma_c^{pred} = 0.90$ to 90.4% at $m = m_c^{pred} + 3\sigma_c^{pred} = 1.54$. These results indicate that these two regions possess the highest earthquake detection quality. Sichuan-Yunnan and California perform slightly worse. In Sichuan-Yunnan, the completeness probabilities increase from 68.3% at $m = m_c^{pred} + \sigma_c^{pred} = 1.48$ to 84.0% at $m = m_c^{pred} + 3\sigma_c^{pred} = 2.26$, whereas in California, probabilities range from 62.0% at $m_c^{pred} + \sigma_c^{pred} = 1.37$ to 82.7% at $m_c^{pred} + 3\sigma_c^{pred} = 2.19$. The poorest observational completeness is found in northern Xinjiang and New Zealand. In northern Xinjiang, the completeness probabilities are 51.3%, 55.0%, and 58.6% at $m_c^{pred} + \sigma_c^{pred} = 1.86$, $m_c^{pred} + 2\sigma_c^{pred} = 2.25$, and $m_c^{pred} + 3\sigma_c^{pred} = 2.64$, respectively. In New Zealand, the corresponding values are 53.3%, 60.8%, and 67.6% at $m_c^{pred} + \sigma_c^{pred} = 2.46$, $m_c^{pred} + 2\sigma_c^{pred} = 2.85$, and $m_c^{pred} + 3\sigma_c^{pred} = 3.24$. The limited performance in northern Xinjiang is primarily due to the sparse station coverage (e.g., Li *et al.*, 2023), whereas in New Zealand it is attributed to the high proportion of earthquakes occurring at depths exceeding 30 km (Petersen *et al.*, 2011). It should be noted that the derived results are subject to uncertainties inherited from parameter estimation, which can be quantified via error propagation throughout the analysis.

Asymmetry in the transition from incompleteness to completeness

The estimated ζ^{pred} values for GR_{AE} are smaller than 1 across all six regions (Table 1; also see Wang *et al.*, 2025), reflecting both regional variability in detection completeness and asymmetry in the transition around m_c^{pred}. Specifically, $\zeta^{pred} < 1$ implies a slower convergence to the linear GR law for $m \geq m_c^{pred}$, relative to the suppression of earthquakes with magnitudes below m_c^{pred}. Beijing–Tianjin–Hebei (0.56 ± 0.11), southeastern coastal (0.51 ± 0.10), and California (0.46 ± 0.07) exhibit relatively larger ζ^{pred} values with small uncertainties. This slower transition in the $m \geq m_c^{pred}$ region where earthquakes are more numerous may explain the smaller uncertainties in the parameter estimations. Sichuan–Yunnan (0.39 ± 0.72) exhibits a moderate ζ^{pred} value with the largest uncertainties. The rather large estimation uncertainty is influenced by statistics, as there is less data to constrain the faster transition in the domain $m \geq m_c^{pred}$. New Zealand (0.24 ± 0.16) and northern Xinjiang (0.11 ± 0.20) obtain the lowest ζ^{pred},

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indicating a very gradual completeness transition. At magnitude $m = m_c^{pred} + 3\sigma_c^{pred} = 2.64$ and 3.24 for northern Xinjiang and New Zealand, respectively, approximately 41.4% and 32.4% of earthquakes are still estimated to be missing.

These results demonstrate the importance of this class of probabilistic models in which m_c is no longer considered a strict cutoff that abruptly separates the complete from the incomplete regions of the CCFMD. Our result challenges the common assumption that a fixed completeness magnitude is adequate, showing instead that catalog incompleteness is more gradual since larger earthquakes can still be missing. Overall, our results indicate that GR_{AE} provides a more accurate, informative, and reliable estimates of seismological parameters among the models considered.

Robustness and stability of asymptotic *b* value estimates

A relevant consideration in interpreting these regional b^{pred} values shown in the Seismicity characterization across regions: Insights from a multimodel augmented GR framework section is the question of asymptotic representation. As noted, some catalogs (e.g., California, New Zealand) do not contain the largest possible events for their respective regions. Our gReLU-based framework addresses this by design: it estimates the asymptotic slope parameter from the observed crossover and curvature in the distribution, analogous to how finite-size scaling in statistical physics recovers true critical exponents from subsystem data (Barber, 1983; Privman, 1990; Cardy, 1996). Therefore, the b^{pred} values in Table 1 and Figure 5 are not merely local slopes over the sampled magnitude range but are inferred estimates of the scaling parameter that would characterize the complete distribution. This raises a direct question: Does our framework make the estimated b^{pred} values conceptually comparable to b values calculated by other approaches that aim to measure the fundamental scaling exponent, even when those approaches use different magnitude ranges or completeness criteria?

Here, we compare b value estimates from two commonly used methods: the MLE (Aki, 1965) and the b-more-positive estimator (Lippiello and Petrillo, 2024). The b-more-positive method is an enhanced version of the original b-positive estimator (van der Elst, 2021). Both leverage positive magnitude differences between earthquakes, successive events in b-positive, and selected pairs in b-more-positive, to effectively mitigate incompleteness effects, even when both events fall below the estimated m_c. This is particularly insightful because closely spaced earthquakes are likely subject to similar detection conditions within short-term seismic sequences. The methods rely on the elegant statistical property that the distribution of differences between exponentially distributed random variables remains exponential with the same β value (equivalent to the b value).

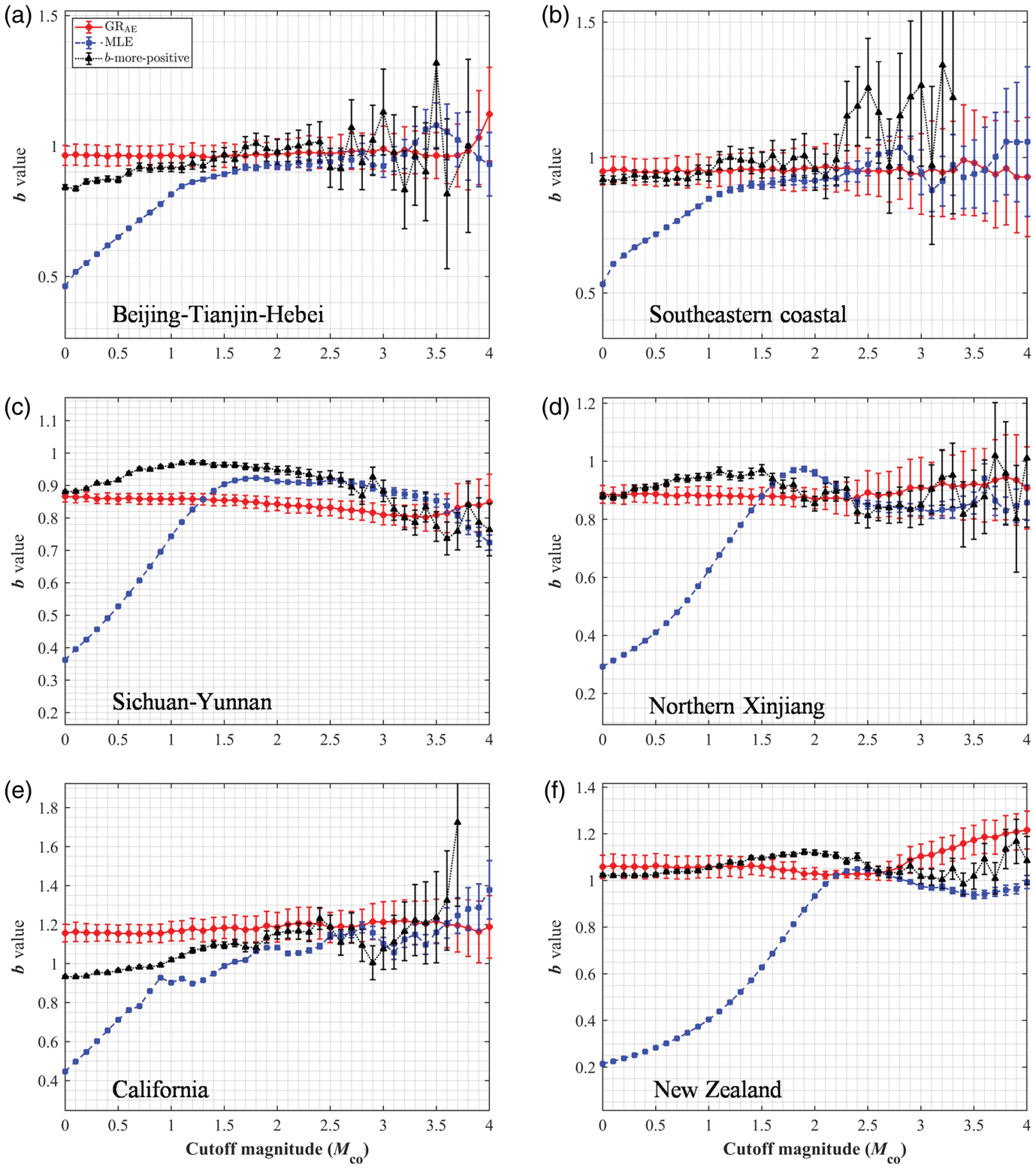
Figure 6 shows the variation of the b value as a function of the catalog cutoff magnitude M_{co} for the six study regions, as estimated by the GR_{AE} model proposed in the present study, MLE, and the b-more-positive method. Error bars indicate $\pm 1\sigma$ standard deviations, with uncertainties derived from 200 bootstrap iterations. The results reveal distinct behavioral patterns across the estimators. For MLE, the estimated b value typically increases with M_{co} when $M_{co} < m_c$, stabilizes when $M_{co} \geq m_c$, and exhibits significant fluctuation when $M_{co} \gg m_c$. This pattern occurs because the FMD deviates from the GR law at lower cutoffs, leading to bias, while sample size dwindles at high cutoffs, increasing uncertainty. The b-more-positive estimator shows a similar overall trend, performing slightly better than MLE at lower magnitudes but with a weaker dependence. However, its fluctuations at the higher-magnitude tail often exceed those of MLE in all six regions, highlighting the high uncertainty inherent to its algorithm in small-sample regimes. In contrast, the b value estimated by our GR_{AE} model demonstrates notable stability and is largely independent of M_{co}; its mean value across the cutoff range approximates a horizontal line. This consistency underscores the reliability and robustness of our physics-informed framework against variations in catalog completeness.

A key consideration in interpreting any earthquake statistics, including the parameters derived from our augmented GR model, is the influence of magnitude scales. Systematic biases and nonlinear scaling between different magnitude types (e.g., M_L, M_D, and M_w) can distort the observed FMD, thereby affecting the fitted completeness parameters and asymptotic b value. Although our model flexibly captures curvature in the empirical distribution, it cannot distinguish between curvature originating from detection limitations and that arising from magnitude-scale inconsistencies. This underscores that the results are intrinsically tied to the homogeneity and accuracy of the input catalog's magnitude system. Importantly, this challenge is not unique to our methodology but is a fundamental, pervasive issue affecting virtually all seismicity-based analyses, from b value estimation and hazard assessment to clustering and forecasting studies. Therefore, rigorous magnitude homogenization remains an essential prerequisite for reliable, comparable seismicity characterization across different networks and regions (e.g., Di Giacomo *et al.*, 2015).

Applicability of the GR_{AE} model to induced seismicity in north-central Oklahoma

To further assess the generality of the proposed framework, we applied the GR_{AE} model to the induced seismicity in north-central Oklahoma during the period 2010–2016, a region characterized by exceptionally dense seismic monitoring and well-documented anthropogenic forcing. This period corresponds to the rapid increase in seismicity associated with large-volume saltwater disposal into the Arbuckle Group, which has been widely interpreted as the primary driver of induced

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earthquakes in Oklahoma (Langenbruch and Zoback, 2016). Seismicity rates increased by nearly three orders of magnitude relative to the pre-2009 background, and the sequence culminated in several moderate-to-large events, including the 2011 M_w 5.6 Prague earthquake and the 2016 M_w 5.8 Pawnee earthquake, the largest instrumentally recorded event in the state. The high detection capability in this region provides an ideal

Figure 6. Variation of the *b* value as a function of catalog cutoff magnitude M_{co}, estimated using the GR$_{AE}$ model proposed in the present study, maximum-likelihood estimation (MLE), and the *b*-more-positive method, respectively, for six study regions: (a) Beijing–Tianjin–Hebei, (b) the southeastern coast, (c) Sichuan–Yunnan, (d) northern Xinjiang, (e) California, and (f) New Zealand. Error bars indicate $\pm 1\sigma$ standard deviations, with uncertainties estimated from 200 bootstrap iterations.

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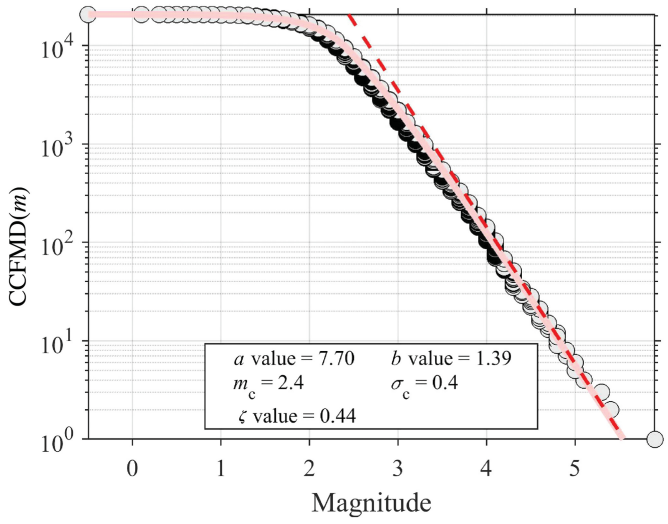


Figure 7. CCFMD of induced seismicity in north-central Oklahoma from 1 January 2010 to 31 December 2016, together with the fitted CCFMD(m) derived using the GR_{AE} model (pink solid line). The red dashed line shows the linear GR law representing the asymptotic behavior on the large-magnitude side. Earthquake data are from Langenbruch and Zoback (2016).

test case for evaluating whether the augmented GR framework can meaningfully characterize induced seismicity beyond tectonic settings.

Fitting the GR_{AE} model to the Oklahoma catalog yields parameter estimates of $a = 7.70$, $b = 1.39$, $m_c = 2.4$, $\sigma_c = 0.4$, and $\zeta = 0.44$ (Fig. 7). The estimated b value is notably larger than unity, consistent with previous studies reporting elevated b values for induced seismicity relative to tectonic backgrounds, reflecting a higher proportion of small-to-moderate events. This estimate agrees closely with the maximum-likelihood b values reported by Langenbruch and Zoback (2016), who found $b \approx 1.33$–1.41 for central and western Oklahoma during the same period. Importantly, the GR_{AE} model achieves this consistency while simultaneously providing an explicit probabilistic description of catalog completeness, rather than relying on a fixed magnitude cutoff.

The inferred completeness structure further highlights the advantages of the augmented formulation for induced seismicity. The relatively small σ_c indicates a sharp but continuous transition from incomplete to complete detection, consistent with the high sensitivity of the Oklahoma seismic network after 2010, whereas the intermediate value of ζ reflects mild asymmetry between the suppression of small-magnitude events and the recovery of the linear GR scaling at larger magnitudes. These results demonstrate that GR_{AE} not only reproduces established statistical properties of induced seismicity but also extends existing approaches by jointly resolving completeness, scaling, and asymmetry within a single, internally consistent model. This suggests that the proposed framework is well-suited for analyzing induced seismicity catalogs and may provide a useful bridge between catalog-based statistical characterization and physics-based forecasting models of injection-related earthquake activity.

Conclusion

We have proposed an augmented GR law incorporating four candidate gReLU(x) functions designed specifically to handle catalog incompleteness. This new model integrates a probabilistic description of catalog completeness with the classical GR law for characterizing earthquake magnitude distributions. The foundational framework and the initial development of the GR_{BS} and GR_{AE} models were established in our related work (Wang *et al.*, 2025). That study evaluated model performance within a traditional framework based on a truncated m_c assumption (incomplete below, complete above). Here, we have extended this perspective by adopting a fundamentally distinct evaluation framework based on the concept of a probabilistic, smoothly varying m_c. We evaluated the four augmented GR models (GR_{SS}, GR_{BS}, GR_{CO}, and GR_{AE}) using synthetic earthquake catalogs—examining both correct model specification (where the fitting model matches the data-generating model) and model mis-specification scenarios—as well as observed earthquake catalogs collected since 2009 from six study regions. Model performance was systematically assessed based on parameter accuracy and uncertainty, goodness-of-fit metrics, and statistical p values. The GR_{AE} model consistently demonstrated superior robustness and reliability across all tests, confirming it as the preferred choice for general application. The strength and physical consistency of this framework are further substantiated from multiple perspectives within this study. This includes establishing the novel concept of magnitude-dependent detection probability, conducting a comprehensive comparison of b value estimates from GR_{AE} against previous methods (MLE and the b-more-positive estimator), and successfully applying the model to a high-resolution catalog of induced seismicity in Oklahoma. These analyses collectively affirm the framework's robustness in characterizing both the physical processes of seismicity and the response characteristics of seismic networks.

Our proposed augmented GR law offers a physically meaningful representation of the gradual transition between fully detected (complete) and partially detected (incomplete) portions of earthquake catalogs. Unlike traditional methods, which rely on a fixed, hard cutoff completeness magnitude m_c, this model interprets catalog completeness as a probabilistic concept characterized by a transition magnitude m_c and a transition-width parameter σ_c. This formulation provides several advantages, especially in situations where the strict cutoff assumption inadequately reflects the complexities and inherent uncertainties associated with real seismic data. Specifically, the augmented GR law accounts for the decreasing probability of detecting smaller earthquakes below m_c, as well as the nonzero

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probability of missing larger events above m_c. The novel enhanced GR model offers distinct advantages: (1) it provides a unified framework that naturally integrates catalog completeness and MFD within a single, coherent formulation; (2) its parameters possess clear physical interpretations and intuitive statistical meanings; and (3) it effectively captures the nonlinear and asymmetric characteristics of earthquake detection probabilities, accurately reflecting real-world phenomena such as rapid detection probability decay and significant asymmetry at lower magnitudes, as commonly observed in empirical earthquake catalogs.

The robustness of the augmented GR approach under heterogeneous completeness conditions has been systematically validated using synthetic tests (Wang *et al.*, 2025). In that study, catalogs with strongly varying m_c^{true} across spatial (or temporal) grids were generated. The results demonstrated that, given a sufficiently large catalog size and fine grid resolution, the GR_{BS} and GR_{AE} models reached the regional mean m_c^{true}, whereas most traditional estimators persistently underestimated it. This performance gap arises because deviations from the GR law at lower magnitudes are often attributable to short-term aftershock incompleteness or short-term network detection incompleteness (Kagan, 2004; Woessner and Wiemer, 2005; Lolli and Gasperini, 2006), phenomena that lead to dynamic variations in m_c across both time and space. Conventional methods, which rely on a single, fixed completeness threshold, fail to adequately capture this variability. In contrast, our probabilistic framework explicitly accounts for such heterogeneity, enabling it to provide reliable m_c estimates for any user-specified completeness level (e.g., 90%, 95%, or 99%), thereby outperforming traditional approaches.

The findings summarized in Figures 5 and 7 strongly support a paradigm shift away from the long-held view of a universal b value constant of 1.0. Our results demonstrate systematic and statistically significant deviations from $b = 1$ across different tectonic environments, with values ranging from below 0.9 in compressional settings to above 1.1 in extensional or transform regions. This pattern directly aligns with and reinforces a substantial body of established literature. As we and others have noted, extensive research confirms the spatiotemporal variability of the b value (e.g., Gulia and Wiemer, 2019; Lombardi, 2023; Taroni and Carafa, 2023), and its real-time estimation without assuming unity is now operational (e.g., Spassiani *et al.*, 2023; Li and Luo, 2024). Crucially, our observed correlation between b value and tectonic regime reaffirms the seminal global pattern documented by Schorlemmer *et al.* (2005) and further explored in subsequent studies (e.g., Gulia and Wiemer, 2010; Petruccelli *et al.*, 2019). Notably, this variability extends to fundamental physical interpretations, as demonstrated by Pisarenko and Sornette (2003), who reported a highly significant difference between midocean ridges ($b \approx 1.509$) and other zones ($b \approx 1.00$), linking it to contrasting rupture healing mechanisms.

Therefore, the assumption of a universal b value is not merely an oversimplification but may lead to biased inputs for seismic hazard assessment, particularly in regions where the true value deviates significantly from 1. Our advocacy for region-specific b values is thus grounded in this well-documented heterogeneity. However, reliably estimating these regional values remains a challenge. In data-sparse regions (common in induced, volcanic, or historical seismicity), standard techniques can yield biased estimates (e.g., Geffers *et al.*, 2022), making a simple empirical derivation from small catalogs as problematic as assuming a default value. Our physics-informed approach addresses this dual challenge. By integrating a probabilistic description of completeness within the GR law, our augmented framework provides a stable method to estimate b values that are robust to data limitations and catalog incompleteness. The consistency of our calibrated estimates with traditional methods across multiple regions suggests they capture fundamental differences in the underlying physics, such as stress state and fault heterogeneity, rather than methodological artifacts. This provides a more reliable path forward, moving beyond both simplistic universal assumptions and potentially misleading empirical estimates derived from limited data, thereby improving the physical basis for seismic hazard assessment worldwide.

Finally, by challenging the traditional assumption of absolute completeness above a fixed magnitude threshold, the augmented GR law provides a flexible, probabilistic framework capable of bridging conventional cutoff-based models and probabilistic completeness approaches. This flexibility is particularly advantageous for applications requiring detailed characterizations of data completeness. For example, our model can serve as a robust null hypothesis for identifying "dragon-king" earthquakes—extreme events that stand apart from typical frequency–magnitude relationships—without necessitating the specification of arbitrary truncation points (e.g., Li, Sornette, Wu, and Li, 2025; Li and Sornette, 2025). Moreover, integrating the augmented GR model into statistical seismicity frameworks such as the epidemic-type aftershock sequence (ETAS) model (e.g., Ogata, 1988, 1998, 2024; Ogata and Zhuang, 2006; Nandan *et al.*, 2021, 2022); will allow catalog quality assessment to be jointly inferred through parameters m_c and σ_c. Including previously discarded small earthquakes—appropriately weighted through these two parameters—can enhance the quality of statistical estimations of ETAS model parameters, such as productivity and branching ratios (Li, Sornette, Wu, *et al.*, 2025), and even help estimate the minimum magnitude capable of triggering aftershocks (m_0; Sornette and Werner, 2005; Zaccagnino *et al.*, 2025). Overall, this innovative and practical augmented GR framework holds substantial promise for advancing our understanding of seismicity and improving the performance of statistical–physical earthquake models, thereby contributing to improved earthquake forecasting. Future work will also focus on addressing the issue of tapering and potential outliers in the

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GR distribution at very large magnitudes (Sornette and Sornette, 1999; Kagan, 2002a,b; Li, Sornette, Wu, and Li, 2025; Li and Sornette, 2025), which requires the development of a new research framework to integrate the asymptotic behavior of the proposed models with these characteristics of magnitude distributions at extreme values.

Incorporating small earthquakes through probabilistic weighting provides a principled alternative to rigid completeness-threshold approaches and has important implications for seismicity forecasting. Although the use of probabilistic weights explicitly acknowledges uncertainty in event detectability, it does not necessarily increase forecast uncertainty. On the contrary, by retaining information from sub-m_c events, albeit downweighted, this approach preserves valuable constraints on the underlying seismicity process, including background rate variations, temporal clustering, and short-term triggering behavior. Discarding such events entirely reduces the effective information content of the catalog and forces forecasts to rely on sparser data, which can increase epistemic uncertainty. Probabilistic inclusion enlarges the effective dataset while transparently accounting for observational limitations, thereby improving parameter estimation and potentially enhancing forecast stability and skill. A systematic quantification of these effects on prospective earthquake forecasts represents a natural and important direction for future work.

Data and Resources

The California catalog is sourced from the Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat; U.S. Geological Survey, 2017), accessible at https://earthquake.usgs.gov/data/comcat/ (last accessed December 2023). For the catalog in New Zealand, the data are obtained from the GeoNet Earthquake Catalog of New Zealand (GNS Science, 2022), accessible at https://quakesearch.geonet.org.nz/ (last accessed December 2023). We acknowledge the New Zealand GeoNet program and its sponsors EQC, GNS Science, LINZ, NEMA, and MBIE for providing data used in the present study. The data for the four study regions in China are acquired from the China Earthquake Networks Center (CENC) through the internal link provided by the Earthquake Cataloging System at China Earthquake Administration, available at http://10.5.160.18/console/index.action (last accessed December 2023), with a digital object identifier (DOI) of 10.11998/SeisDmc/SN. These data are not publicly available; they can be requested from the CENC. The earthquake catalog for the four study regions in China, covering events with magnitudes ≥ 0 from 2009 to 2024 (excluding time and location information), along with the MATLAB (www.mathworks.com/products/matlab, last accessed May 2026) codes for algorithms developed in this study, are openly available at Zenodo: doi: 10.5281/zenodo.15797118 (Li, Wang, and Sornette, 2025). The supplemental material for this article includes details of the fitting algorithm for numerical stability and robustness, model performance evaluations on synthetic and empirical catalogs, seismological interpretations of key parameters, supplemental figures (parameter probability density functions [PDFs], cumulative frequency–magnitude distributions [CFMDs], and residual diagnostics), and tables summarizing parameter estimates and performance metrics.

Declaration of Competing Interests

The authors acknowledge that there are no conflicts of interest recorded.

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